

CALEB MANTEY

Software / XR Engineer Tema Metropolitan District, Ghana | manteyleb@gmail.com
[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Versatile Software and XR Engineer with 7+ years of experience in full stack web architecture and immersive xr experiences. Expert in building high-concurrency backends (Ruby on Rails, NestJS) and high-performance XR applications (Unity, Three.js). Winner of the **2023 NEIP Presidential Pitch** (Ghc 200,000 grant) for a simulation-based training innovation . Proven track record in scaling platforms like **besaCare**, supporting 500+ pharmacies and 3,000+ daily medical tests.

TECHNICAL SKILLS MATRIX

- **Fullstack & Backend:** Ruby on Rails, NestJS, Node.js, Python (FastAPI), Java (Spring Boot), PostgreSQL, MySQL, Redis, REST/GraphQL APIs.
- **XR & Game Engineering:** Unity3D (C#), Three.js, React-three-fiber, WebXR, ARCore/ARKit, visionOS, GLSL Shaders, 3D Mesh Optimization (Draco/KTX2) .
- **Frontend & Mobile:** React, Next.js, Angular, Flutter, React Native, TypeScript, Tailwind CSS.
- **AI & Automation:** LangGraph, LangChain, RAG Orchestration, OpenAI API, Vector Databases (Pinecone, Weaviate), RAGAS Evaluation.
- **DevOps & Tools:** Docker, CI/CD (GitHub Actions), AWS, DigitalOcean, Linux, Git, Nginx.

PROFESSIONAL EXPERIENCE

Relu Interactives | Co-Founder & CTO | Accra, Ghana *September 2022 – Present*

- **Secured GH¢ 200,000 (\$20k equivalent grant)** by winning Season 4 of the NEIP Presidential Pitch; leveraged funds to scale industrial simulation products .
- **Developed high-fidelity VR experiences** for various commercial and industry clients in Ghana, utilizing Unity and spatial logic to simulate hazardous environments and rich interactive experiences.
- **Optimized WebXR performance** via Three.js, achieving stable 60 FPS on mobile XR browsers by implementing instanced rendering and asset compression (Draco) .
- **Architected and led development** of XR applications across WebXR, VR headsets, and mobile platforms
- **Directed ReluSpatial** (web-based 3D collaboration platform), focusing on accessibility and real-time interaction
- **Created VR simulations** including "PetSim", a petroleum engineering training rig, used in academic and industrial training

- **Produced digital twins and immersive simulations** for education, engineering, and heritage clients
- **Managed cross-functional teams** across design, development, and deployment

Zenjob | Fullstack & AI Software Developer | Berlin, Germany *July 2024 – May 2025*

- **Engineered an AI-driven matchmaking engine** using LangGraph and Python, resulting in a 25% increase in placement speed for student labor across Germany.
- **Automated operational workflows** via multi-agent AI systems, saving 15+ manual vetting hours weekly and reducing data entry errors.
- **Built production-ready RAG pipelines** with DeepEval guardrails, ensuring 90%+ faithfulness in automated business response systems.

Think-it | Software Engineer | Germany (Remote) *April 2023 – June 2025*

- **Architected large-scale web applications** for climate-tech and health-tech partners, focusing on scalability and third-party API orchestration.
- **Integrated Generative AI and LLMs** into enterprise product workflows, enabling autonomous data analysis for mission-driven global organizations.

Andela / Prosperity Health Ventures | Senior Software Engineer | Global *September 2021 – January 2024*

- **Scaled Backends:** Optimized backend infrastructures for global clients, resolving critical performance bottlenecks and reducing API latency by 30%.
- **Web Games:** Built and deployed WebGL/Three.js games, XR prototypes, and scalable platforms

eGotickets / Encodev Labs | Software Engineer | Tema, Ghana *September 2020 – April 2023*

- **High-Concurrency Ticketing:** Maintained a platform supporting 10,000+ peak concurrent users with 99.99% uptime using Ruby on Rails and Redis.
- **Payment Systems Integration:** Architected secure payment gateway integrations for diverse client applications, handling high-volume financial transactions.
- **CI/CD Leadership:** Automated deployment pipelines using Docker and GitHub Actions, reducing manual deployment time by 60%.
- **Helped in Architecture decisions for "besaCare":** Scaled a telemedicine and diagnostic platform to a network of **500 pharmacies**, processing **2,000-3,000 patient tests daily**.
- **Developed Cross-Platform Mobile Apps:** Built the besaCare patient/provider apps using React, Capacitor and Ruby On Rails, integrating digital health records and chat functionality for consultations.

SokoAerial Robotics | Software & VR Engineer | Accra, Ghana *September 2019 – August 2020*

- **Military VR Research:** Pioneered VR/AR training prototypes for the military, focusing on immersive spatial awareness and tactical simulations.
- **Engineering Mentorship:** Led a team of interns, facilitating onboarding and technical training in React and Unity development.

Stanbic Bank Ghana | Software Engineer | Accra, Ghana *June 2019 – October 2019*¹

- **Enterprise Java Development:** Developed frontend components using Angular and backend logic with Java (Spring Boot) for secure banking portals.

HONORS & LEADERSHIP

- **Overall Winner:** NEIP Presidential Pitch Season 4 (2023) .
- **Animation/Gaming Startup of the Year:** Association of Ghana Startups (2023) .
- **Invited Facilitator:** Society of Petroleum Engineers (SPE) Ghana – Immersive Tech in Energy .
- **Awards:** Most Innovative Student (UMaT), 1st Runner Up HacklabGH Hackathon.

EDUCATION

University of Mines and Technology (UMaT) B.Sc. Geomatic Engineering¹ *Focus: Spatial Data, GIS, and Coordinate Systems (WGS84/UTM).*